

Speed Control of Separately Excited DC Motor using Buck Converter under Varying Load

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Abstract- This project investigates the speed control of a Separately Excited DC Motor using a DC-DC buck converter under varying load torque conditions, simulated in MATLAB software. The study focuses on designing a closed-loop control system to maintain motor speed despite load disturbances. The system incorporates PWM-based voltage regulation and PID speed control. Results demonstrate the effectiveness of the proposed method in maintaining stable speed under dynamic torque variations, with comparisons made to Series and Shunt DC motor configurations.

Keywords- PID control, DC-DC buck converter, DC motor, Speed control, Load torque, Voltage regulation, etc.

Prior research has explored solutions like PWM-based voltage regulation and PID control, with buck converters gaining attention for their efficient voltage step-down capability. Studies demonstrate their potential, but gaps remain in optimizing dynamic response and robustness for real-world industrial loads. This project advances these efforts by: Proposing a PI-controlled buck converter system tailored for Separately Excited DC Motors, validating performance under step load changes (0.5–2.0 Nm) via MATLAB, comparing results with Series/Shunt motors to highlight advantages of separate excitation.

By integrating proven techniques with targeted improvements, this work aims to deliver a practical, simulation-validated solution for industrial speed control challenges.

1. INTRODUCTION

DC motors are widely used in industrial automation, particularly in applications like conveyor belt systems, due to their simplicity, cost-effectiveness, and ease of speed control. However, load variations often lead to speed instability, compromising system performance. To address this challenge, a DC-DC buck converter can regulate the motor's supply voltage, enabling precise speed control even under fluctuating loads. This project investigates a closed-loop control system for a Separately Excited DC Motor driven by a buck converter, evaluating its effectiveness through PSIM simulations under dynamic load conditions.

The ability to maintain consistent motor speed is critical in industrial automation, where speed fluctuations can disrupt processes like material handling or assembly lines. While DC motors offer straightforward control, their speed-torque characteristics make them susceptible to load disturbances. For instance, in conveyor systems, sudden load changes can cause speed dips or overshoots, reducing efficiency. This project focuses on answering: How can a buck converter-based control system compensate for load torque variations while maintaining stable motor speed?

This problem is both practically significant and technically challenging. Industrially, robust speed control enhances energy efficiency and operational precision. However, achieving this requires addressing: Nonlinearities in motor dynamics under load changes, Trade-offs between response speed and stability in control design, Power losses in the converter-motor system.

2. Methods

This section details the design and simulation methodology employed to develop the DC motor speed control system. The system was modeled and simulated using MATLAB/SIMULINK.

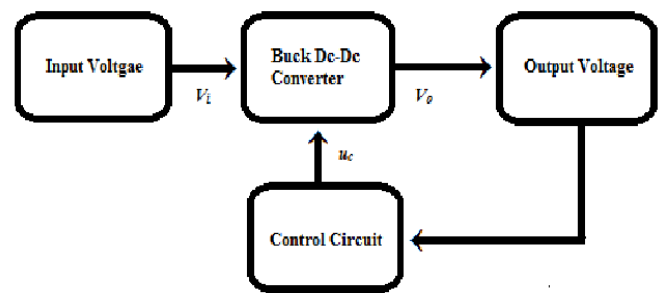


Figure 1: Block Diagram of the system

2.1 System Description

The overall system, implemented in Simulink (Figure 1), consists of the following key stages:

1. **Power Conversion:** A DC-DC buck converter regulates the DC voltage supplied to the motor, enabling control of its armature voltage.

2. Motor Drive: A DC motor converts the electrical energy into mechanical energy to drive the conveyor belt.
3. Load Dynamics: The conveyor belt's load is represented as mechanical torque acting on the motor shaft.
4. Closed-Loop Control: A PID controller adjusts the buck converter's operation (duty cycle) based on the difference between the desired and actual motor speed, ensuring accurate speed regulation.

2.2 Buck Converter Implementation and Design

The buck converter is a DC-DC power converter that steps down a DC input voltage (V_{in}) to a lower DC output voltage (V_{out}). It operates by switching a semiconductor switch (MOSFET) on and off, controlling the average voltage delivered to the load. The key components of the buck converter are:

- Switch (MOSFET): Controls the flow of current.
- Diode: Provides a path for the inductor current when the switch is off.
- Inductor (L): Stores energy and smooths the current flow.
- Capacitor (C): Smooths the output voltage.

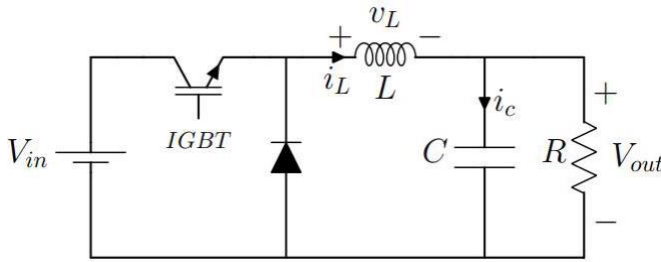


Figure 2: Buck-converter Circuit Diagram

The output voltage of an ideal buck converter is given by:

$$V_{out} = D \cdot V_{in}$$

Where D is the duty cycle, defined as the ratio of the switch's on-time to the switching period.

$$V_0 = 300v, V_s = 400v, R = 100 \text{ ohm}, r = 0.5\%, f = 40 \text{ kHz}$$

$$D = \frac{V_0}{V_s} = \frac{300}{400} = 0.75$$

$$L_{min} = \frac{(1-D)R}{2f} = 3.125 \times 10^{-4} H$$

$$L = 1.25 * L_{min} = 3.90625 \times 10^{-4} H$$

$$C = \frac{1-D}{8Lr f^2} = \frac{0.25}{8(3.90625 \times 10^{-4} H)(0.005)(40,000^2)} = 10^{-5} \text{ farad}$$

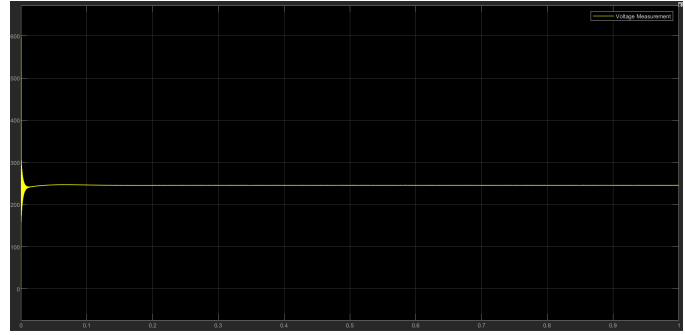


Figure 3: Output Voltage

2.3 DC Motor Implementation

For this application, a separately excited DC motor is chosen due to its superior speed control characteristics compared to series and shunt motors, as discussed in the Introduction. Series motors have poor speed regulation, and while shunt motors are better, separately excited motors offer independent control of field flux, allowing for precise speed control under varying load conditions. The DC motor is modeled as a separately excited motor. The separately excited DC motor consists of:

- Armature Resistance (R_a)
- Armature Inductance (L_a)
- Field current (I_f)

Parameter	Value
Armature Resistance (R_a)	0.6Ω
Armature Inductance (L_a)	$0.012H$
Torque Constant (K_t)	$0.95Nm/A$
Inertia(J)	$0.05kg.m^2$
Viscous Friction (B_m)	$0.005Nms$
Columb Friction (T_c)	$0Nm$
Desired Speed (ω)	$200rad/sec$
Desired Torque (T_e)	$100Nm$

- To Calculate the Required Armature Current

The electrical torque is related to the armature current I_a by the following equation:

$$T_e = K_t \cdot I_a$$

Where:

- $T_e = 100 \text{ Nm}$
- $K_t = 0.95 \text{ Nm/A}$

Solving for I_a :

$$I_a = \frac{T_e}{K_t} = \frac{100}{0.95} = 105.26A$$

So, the armature current I_a required to achieve 100 Nm of torque is **105.26 A**.

- To Calculate the Required Armature Voltage (V_a)

Using the motor speed formula:

$$\omega = \frac{V_a - R_a I_a}{K_t}$$

Rearranging to solve for V_a :

$$V_a = \omega \cdot K_t + R_a \cdot I_a$$

Where:

- $\omega=200$ rad/sec
- $R_a=0.6 \Omega$
- $I_a=105.26A$
- $K_t=0.95$ Nm/A

Substituting the values:

$$V_a = 200 \cdot 0.95 + 0.6 \cdot 105$$

$$V_a = 190 + 63.16 = 253.16V$$

So, the required armature voltage V_a to achieve 100 Nm of torque and a speed of 200 rad/sec is **253.16 V**.

Variable	Value
Desired Torque	100Nm
Desired Speed	200rad/sec
Required Armature Voltage	253.16V
Required Armature Current	105.26A

2.4 Control System Implementation

The main control method involves using a PID controller to adjust the PWM duty cycle based on the motor's speed.

- Proportional (P): This term adjusts the control output based on the current error, i.e., the difference between the desired speed and the actual motor speed.
- Integral (I): The integral term compensates for accumulated past errors, eliminating any steady-state error over time.
- Derivative (D): The derivative term anticipates future errors, helping to dampen the system's response and reduce overshoot or oscillations.

The PID controller receives feedback from the motor's speed sensor, compares it with the desired speed, and generates a PWM signal accordingly to adjust the converter's duty cycle. This allows the motor speed to remain constant despite fluctuations in load torque.

A closed-loop PID controller is used to regulate the speed of the DC motor. The PID controller calculates the error between the reference speed (ω_{ref}) and the actual motor speed (ω_m) and generates a control signal to adjust the duty cycle of the buck converter. The PID controller equation is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

The Proportional-Integral-Derivative (PID) controller parameters, namely K_p (proportional gain), K_i (integral gain), and K_d (derivative gain), were tuned to achieve the desired system performance. The tuning process was carried out using the [e.g., 'MATLAB PID Tuner' tool / or 'trial and error' method] to meet performance objectives such as fast rise time, minimal overshoot, and zero steady-state error.

After completing the tuning process, the final PID gains were determined as:

- $K_p=80$ (Proportional gain)
- $K_i=800$ (Integral gain)
- $K_d=0$ (Derivative gain)

These values indicate that the controller relies primarily on proportional and integral actions. A relatively high integral gain K_i selected to eliminate the steady-state error efficiently, while the derivative gain K_d was set to zero, suggesting that derivative action was either unnecessary or introduced instability or noise in the system. As a result, the system operates essentially as a PI controller rather than a full PID controller, providing stable and satisfactory dynamic and steady-state responses under varying load conditions.

2.5 Simulation Methodology

The performance of the proposed DC motor speed control system was evaluated through simulations in MATLAB/Simulink. The simulation was configured with a Stop Time of 1 second, during which key system variables, such as motor electrical torque, speed, and voltage, was monitored and recorded using Scope blocks. The simulation also incorporated a Random Number block to introduce random load disturbances, providing a dynamic testing environment for the control system.

The system architecture centers around a buck converter used to regulate the voltage supplied to the DC motor. A Pulse Generator block produces the PWM signal, which is modulated through a Product block and used to control an IGBT/Diode switching device. Energy storage and smoothing are handled by a Series RLC Branch modeling the inductor and capacitor. A freewheeling diode is used to allow continuous current flow when the IGBT is off, maintaining proper buck converter operation.

A PID Controller is implemented to adjust the duty cycle of the PWM signal based on the difference between the desired and actual motor speeds. The PID output is passed through a Saturation block to ensure the duty cycle remains within permissible limits. The motor's speed reference is provided by a Constant block, and the actual motor speed feedback is derived from the motor's output,

completing the closed-loop speed control.

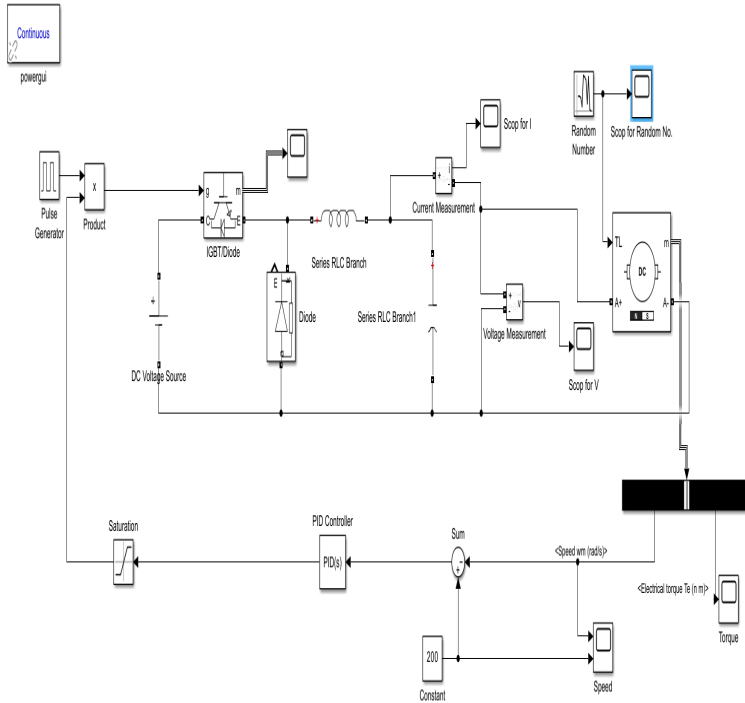


Figure 4: Simulink model of the complete system

The DC motor model used in the simulation represents both its electrical (armature resistance and inductance, back EMF) and mechanical (inertia, friction, and load torque) dynamics. As the motor load varies due to the randomized disturbance, the PID controller dynamically adjusts the duty cycle of the buck converter to maintain the motor speed close to the set reference.

Supporting Figures

Supporting figures are included in the report to provide a clear overview of the system:

- Figure 1: Block Diagram of the system
- Figure 2: Buck-converter Circuit Diagram
- Figure 3: Output Voltage
- Figure 4: Simulink model of the complete system
- Figure 5: Graph of Random Variable for T_L
- Figure 6: Graph of Output current under varying T_L
- Figure 7: Graph of Output Electrical Torque
- Figure 8: Motor speed versus time graph

Simulation Results and Analysis

The system was tested under different load conditions to assess its performance:

- **Steady-State Performance:**
The motor was able to maintain the reference speed with minimal steady-state error despite the presence of random load disturbances.

- **Transient Response:**

The system demonstrated quick recovery following sudden changes in load torque, stabilizing to the reference speed within a short time without significant overshoot or oscillations.

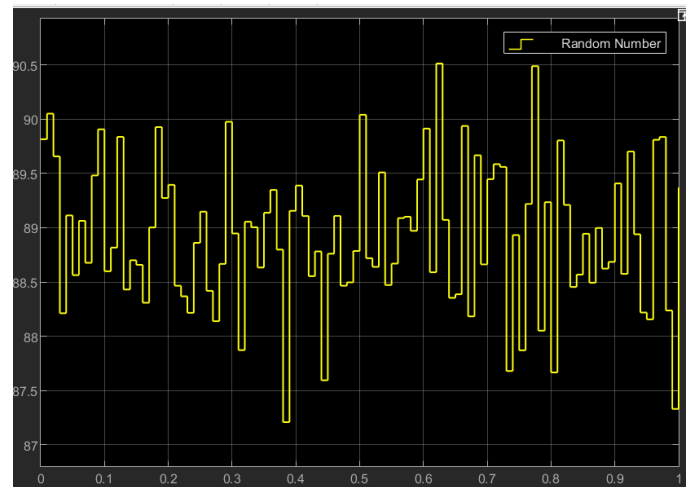


Figure 5: Graph of Random Variable for T_L

The figure 5 illustrates the random load torque disturbance applied to the DC motor during the simulation. The Random Number block was configured with the following parameters:

- **Mean:** 90
- **Variance:** 0.5
- **Seed:** 456

This disturbance simulates real-world load variability and is used to evaluate the robustness and dynamic performance of the speed control system under varying load conditions.

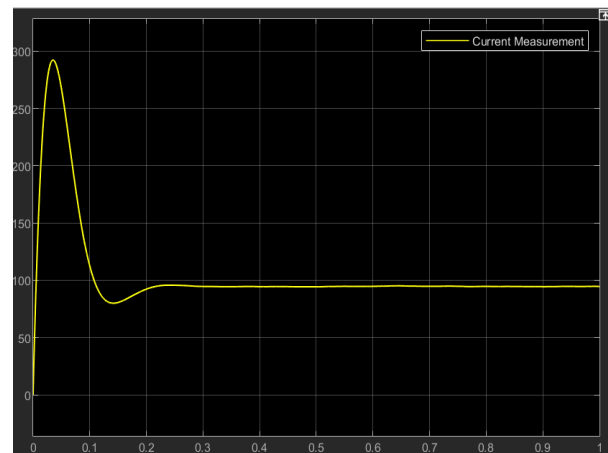


Figure 6: Graph of Output current under varying T_L

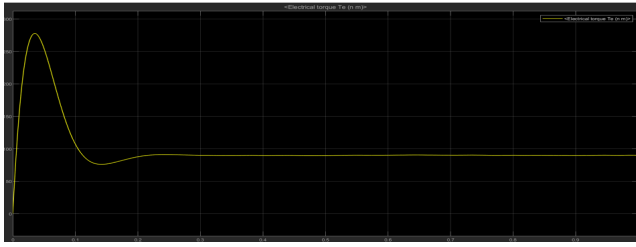


Figure 7: Graph of Output Electrical Torque

The two figures 6 & 7 shows the dynamic response of the motor current and electrical torque following the application of a random load disturbance. The response characteristics are as follows:

- **Initial Overshoot:** Both the armature current and torque exhibit a sharp peak at the start of the simulation due to the initial acceleration demand of the motor.
- **Settling Behavior:** After the initial overshoot, the current and torque settle smoothly toward their steady-state values.
- **Steady-State Response:** Throughout the remainder of the simulation, both the current and torque maintain a nearly constant value, indicating that the control system successfully compensates for load variations and maintains stable motor operation.

This response demonstrates the effectiveness of the PID-controlled buck converter in rapidly stabilizing the motor after disturbances while minimizing oscillations and ensuring consistent performance under variable load conditions.

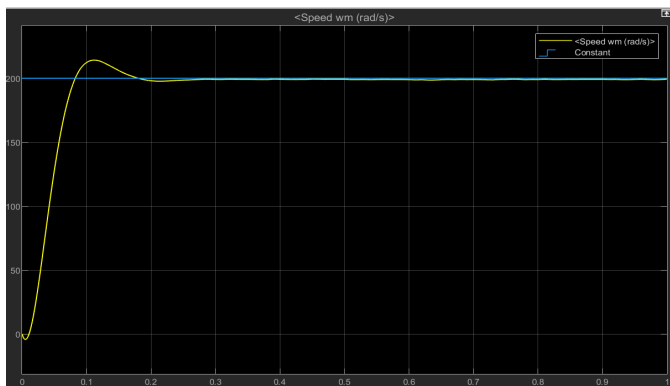


Figure 8: Motor speed versus time graph

The figure 8 presents the dynamic response of the DC motor's speed compared to the reference speed command. The response characteristics are as follows:

- **Initial Acceleration:** At the start of the simulation, the motor rapidly accelerates to reach the desired reference speed.

- **Overshoot:** A small overshoot is observed shortly after startup, which is characteristic of PID-controlled systems tuned for fast response.
- **Settling Time:** The motor speed quickly stabilizes and aligns with the reference speed after a brief transient period.
- **Steady-State Performance:** Once settled, the motor maintains a nearly constant speed, effectively tracking the reference value despite the presence of random load disturbances.

This behavior demonstrates that the speed control system, using a PID-controlled buck converter, successfully achieves fast and stable tracking of the desired motor speed under varying load conditions.

Conclusion

The main objective of this project was to design and simulate a speed control system for a separately excited DC motor that could be effectively applied to conveyor belt systems, where stable and precise speed regulation is essential. A DC-DC buck converter, controlled by a PID controller, was used to regulate the motor's speed under varying load torque conditions. The system was modeled and simulated in MATLAB/Simulink, utilizing a closed-loop feedback control strategy. A Random Number block introduced load torque disturbances to emulate realistic operating environments and to evaluate the dynamic response of the system.

Simulation results demonstrated that the PID-controlled buck converter achieved rapid speed stabilization, minimal overshoot, and excellent steady-state performance, even in the presence of unpredictable load variations. Compared to traditional open-loop systems, the proposed control method offered improved robustness, faster recovery from disturbances, and higher precision, making it ideal for conveyor belt applications where consistent speed is critical for operational efficiency.

Reference

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